

Women's Specialists of Clear Lake

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Tay-Sachs Disease Carrier Screening

Understanding carrier screening and your pregnancy options. Screening is optional — its purpose is to give you information for your decisions.

WHAT IS TAY-SACHS DISEASE?

- A rare **inherited** disorder that affects the brain and nervous system
- Caused by changes in a gene called **HEXA**
- The body can't make enough of an enzyme that breaks down fatty substances in nerve cells
- The most common form begins in **infancy**; rarer forms start in childhood or adulthood

THE ODDS IF BOTH PARENTS CARRY IT

- 25% (1 in 4)** chance the child will have Tay-Sachs disease
- 50% (1 in 2)** chance the child will be a healthy carrier
- 25% (1 in 4)** chance the child will neither have it nor carry it
- These odds are the **same with every pregnancy**

HOW SCREENING IS DONE

- Genetic testing** looks for changes in the HEXA gene
- Enzyme testing** measures hexosaminidase A activity
- The right test depends on ancestry, family history, and pregnancy status
- If pregnant or on birth control pills, a **blood-cell (leukocyte)** enzyme test is preferred
- A negative result greatly lowers — but never fully removes — carrier risk

WHAT DOES BEING A CARRIER MEAN?

- A carrier has **one working** and **one changed** copy of the HEXA gene
- Carriers are **healthy** and do **not** develop Tay-Sachs disease
- The condition is passed down in an **autosomal recessive** pattern
- A child must inherit a changed gene from **both parents** to be affected

WHO SHOULD CONSIDER SCREENING?

- ✓ Anyone planning or already in a pregnancy may consider it
- ✓ Especially **Ashkenazi Jewish, French Canadian, Cajun, or Pennsylvania Dutch** ancestry
- ✓ A **family history** of Tay-Sachs or a known HEXA change
- ✓ Carriers occur in **all backgrounds** — testing may be offered to anyone

IF YOUR SCREEN IS POSITIVE

- A positive result does **not** mean you have Tay-Sachs disease
- If one partner is a carrier, the **other partner should be tested**
- If **both** are carriers, genetic counseling explains your options
- Options include prenatal testing (**CVS/amnio**), **IVF** with genetic testing, or **donor egg/sperm**
- Close **biological relatives** may also be carriers and could consider testing

QUESTIONS TO ASK YOUR DOCTOR

- ☐ Should I be screened for Tay-Sachs disease?
- ☐ Which type of testing is right for me?
- ☐ Has my reproductive partner been tested?
- ☐ What does my result mean for this or a future pregnancy?
- ☐ Would genetic counseling be helpful for us?

♥ Screening Is Your Choice

Carrier screening is completely optional. Its only purpose is to give you clear information so you can make reproductive decisions that fit your own values and goals. Whatever you decide, we're here to answer your questions and support you. Please reach out to our office anytime.